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Date 18.02.2022

Reference	Application No./Patent No.
EUR1637WEP	20881549.8 - 1001 / 3893188 PCT/KR2020012685
Applicant/Proprietor	
Ftlab Co., Ltd.	

Communication

The extended European search report is enclosed.

The extended European search report includes, pursuant to Rule 62 EPC, the supplementary European search report (Art. 153(7) EPC) and the European search opinion.

Copies of documents cited in the European search report are attached.

☒ 0 additional set(s) of copies of such documents is (are) enclosed as well.

Refund of the search fee
If applicable under Article 9 Rules relating to fees, a separate communication from the Receiving Section on the refund of the search fee will be sent later.

Should you wish to further prosecute this application in the examination phase, your attention is drawn to the provisions of Rule 70a EPC. An invitation to respond to the extended European search report will be issued shortly (R. 70(2) EPC).

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SUPPLEMENTARY EUROPEAN SEARCH REPORT

Application Number
EP 20 88 1549

DOCUMENTS CONSIDERED TO BE RELEVANT

Category	Citation of document with indication, where appropriate, of relevant passages	Relevant to claim	CLASSIFICATION OF THE APPLICATION (IPC)
X	KR 101 922 211 B1 (BETTERLIFE CO LTD) 26 November 2018 (2018-11-26) * paragraphs [0014], [0024], [0031], [0033], [0034], [0116], [0129], [0134], [0135], [0137], [0143], [0161], [0167], [0208]; figures 1,4,5 *	1-19	INV. G01T1/178 G01T7/00 G06Q50/26 G06Q50/10 F24F11/00 G06Q50/30
X	KR 2018 0057549 A (JOHNSON & JOHNSON VISION CARE [US]) 30 May 2018 (2018-05-30) * paragraphs [0012] - [0014], [0018], [0036], [0046], [0054], [0090]; figures 1,2a,b *	1	F24F110/68
A	JP 4 551430 B2 (HITACHI LTD) 29 September 2010 (2010-09-29) * paragraphs [0034], [0045], [0077], [0084] *	1,12,13	

TECHNICAL FIELDS SEARCHED (IPC)

G01T
G01V

The supplementary search report has been based on the last set of claims valid and available at the start of the search.

Place of search	Date of completion of the search	Examiner
The Hague	10 February 2022	Van Ouytsel, Krist'l

CATEGORY OF CITED DOCUMENTS

X : particularly relevant if taken alone
Y : particularly relevant if combined with another document of the same category
A : technological background
O : non-written disclosure
P : intermediate document

T : theory or principle underlying the invention
E : earlier patent document, but published on, or after the filing date
D : document cited in the application
L : document cited for other reasons

& : member of the same patent family, corresponding document

EPO FORM 1503 03.82 (P04C04)

ANNEX TO THE EUROPEAN SEARCH REPORT ON EUROPEAN PATENT APPLICATION NO. EP 20 88 1549

This annex lists the patent family members relating to the patent documents cited in the above-mentioned European search report. The members are as contained in the European Patent Office EDP file on
The European Patent Office is in no way liable for these particulars which are merely given for the purpose of information.

10-02-2022

Patent document cited in search report	Publication date	Patent family member(s)	Publication date
KR 101922211 B1	26-11-2018	CN 112005269 A	27-11-2020
		KR 101922211 B1	26-11-2018
		US 2020378938 A1	03-12-2020
		WO 2019212139 A1	07-11-2019
KR 20180057549 A	30-05-2018	AU 2017258908 A1	07-06-2018
		BR 102017024886 A2	30-10-2018
		CA 2986134 A1	21-05-2018
		CN 108089000 A	29-05-2018
		EP 3324320 A1	23-05-2018
		JP 2018100957 A	28-06-2018
		KR 20180057549 A	30-05-2018
		RU 2017139862 A	16-05-2019
		SG 10201709261R A	28-06-2018
		TW 201831136 A	01-09-2018
		US 2018144092 A1	24-05-2018
JP 4551430 B2	29-09-2010	JP 4551430 B2	29-09-2010
		JP 2009063510 A	26-03-2009
		US 2009070070 A1	12-03-2009

Information on Search Strategy - Pilot phase (see OJ 2015, A86)
The type of information contained in this sheet may change during the pilot for improving the usefulness of this new service.

Application Number

EP 20 88 1549

TITLE: RADON MANAGEMENT SYSTEM USING RADON DETECTOR

APPLICANT: Ftlab Co., Ltd.

IPC CLASSIFICATION: G01T1/178, G01T7/00, G06Q50/26, G06Q50/10, F24F11/00, G06Q50/30, F24F110/68

EXAMINER: Van Ouytsel, Krist'l

CONSULTED DATABASES:

CLASSIFICATION SYMBOLS DEFINING EXTENT OF THE SEARCH:

IPC:

CPC: G01T1/167, G01T1/178, G01T1/16, G01T1/169, G01V5/0083, G01T7/00

FI/F-TERMS:

KEYWORDS OR OTHER ELEMENTS FEATURING THE INVENTION:
A radon management system, comprising: at least one of radon detectors installed in a specific space indoors or outdoors, and configured to detect in real time alpha particles present in the space, output a predetermined alpha particle detection signal, count for a measurement time preset based on the signal to calculate an alpha particle concentration value, and transmit unique device identification information together with the calculated value; and a radon management server configured to collect the device identification information together with the concentration value transmitted from each radon detector during a preset measurement period, calculate and quantify an average value, convert it into a database, store and manage and compare the value transmitted in real time and the average value of the previously stored concentration values with each other, calculate an amount of change and generate radon generation event information data according to the amount of change.

EPO FORM P04A42

The examination is being carried out on the **following application documents**

Description, Pages

1-44 filed with entry into the regional phase before the EPO

Claims, Numbers

1-19 filed with entry into the regional phase before the EPO

Drawings, Sheets

1/2, 2/2 filed with entry into the regional phase before the EPO

In addition to the findings of the International Search Authority, following further arguments are provided.

Reference is made to the following documents; the numbering will be adhered to in the rest of the procedure.

1. Novelty

The present application does not meet the requirements of Article 52(1) EPC because the subject-matter of claims 1-3, 9, 19 are not new within the meaning of Article 54(1) and (2) EPC.

1.1 Radon management system claim 1

Document D1 (fig.s 1,5) discloses:

A radon management system (fig.1, par.s [14,24]) using a radon detector (fig.s 1,4, ref. 300), the radon management system comprising:
at least one of radon detectors (ref.300, par.s [24,143]) installed in a specific space indoors or outdoors (par.s [14,143]), and configured to detect in real time alpha particles present in the specific space (par.[116]), output a predetermined alpha particle detection signal, count for a measurement time preset based on the output alpha particle detection signal to calculate an alpha particle concentration value (par.[14,31]), and transmit unique device identification information together with the calculated alpha particle concentration value (par.[116]); and
a radon management server (fig.1, ref.500, par.[116]) configured to collect the unique

device identification information (par.[129]) together with the alpha particle concentration value transmitted from each radon detector during a preset measurement period, calculate and quantify an average value of the collected alpha particle concentration values (par.s [14,31,134]), convert the average value into a database (DB) for each radon detector ([14]), store and manage the average value (par.[129]), compare the alpha particle concentration value transmitted in real time from each radon detector and the average value of the previously stored alpha particle concentration values with each other (par.s [24,33,143]), calculate an amount of change in the alpha particle concentration value, and generate radon generation event information data according to the amount of change in the calculated alpha particle concentration value (par.s [33,143,167]).

1.2 Dependent claims 2-19

claim 2: D1 (par.s [33,143]) also discloses the preset reference amount of change.

claim 3: D1 (par.s [135,145,146]) also discloses the radon management server to transmit data to an external terminal through a communication network.

claim 9: D1 (par.s [14,25,33,134,167]) discloses the pattern (graph) determination of the first alpha measurement period.

claim 19: D1 (par.s [129,137,161], fig.s 2,3, ref.s 250,350) also disclose the display of the radiation management information data on an hourly, daily, monthly, yearly basis.

2. Inventive step

The present application does not meet the requirements of Article 52(1) EPC because the subject-matter of claims 1,4-8,10-18 does not involve an inventive step within the meaning of Article 56 EPC.

2.1 Claim 1

Also the teaching of D2 (fig.s 1,2a,b, par.s [12-14:1st line,18,36,46,54,90]) is seen to take away an inventive step of claim 1. Even though a device identification is not explicitly mentioned, given that the location of the patient is registered (par.[90]), and that real-time pattern recognition via a neural network from chemical array data is carried out, the skilled person would arrive at device identification through ordinary experimentation.

For further support, reference is made to D3 (fig.1, par.s [51,90]) detailing device identification for a radon management system.

2.2 Dependent claims

claims 4-8,10-11: taking into account the worker schedule, user mission information or excluding data related to noise when calculating the average concentration value are of course common measures-known in the art.

claims 12-13: D1 (par.s [14,25,34,137]) also discloses radiation mitigation indoors by ventilation. To take into account the user periodic action information in order to control the operation of the radon reduction facility, also is seen to represent but an obvious measure to the skilled person.

Document D3 (par.s [34,45,77,84]) also mentions user actions and measurement devices indicating change or not for the worker, lending further support to the above arguments.

claims 14-15: D1 (par.s [137,167]) also discloses generating a warning message. The claims provide standard situations, common in the art, wherein a warning message or alarm would be issued and thus cannot be seen to contribute to any inventive step.

claims 16-17: though it would appear that the system of D1 measures over somewhat longer times, the purpose is to measure in real time, so that the value of 10 minutes claimed would be arrived at quite automatically by the skilled person, attempting to set and experiment with the system.

claim 18: D1 (par.s [145-146]) also hints at using standard network, computing and storage systems, so that the cloud computing service and encryption options have to be seen to represent standard options to the skilled person for managing server information.

3. Concluding remarks

3.1 It is not at present apparent which part of the application could serve as a basis for a new, allowable claim. Should the applicant nevertheless regard some particular matter as patentable, an independent claim should be filed taking account of Rule 43(1) EPC. The applicant should also indicate how the subject-matter of the new claim differs from the state of the art and the significance thereof, by providing the precise passages relied upon.

It is further noted that the application may not be amended in such a way that it contains subject-matter which extends beyond the content of the application as filed (Article 123(2) EPC).

3.2 The independent claims should further be drafted in the preferred two part form against the prior art of document D1 (Rule 43(1) EPC).

3.3 The features of the claims should furthermore be provided with reference signs placed in parentheses to increase the intelligibility of the claims (Rule 43(7) EPC). This applies to both the preamble and characterising portion (see the Guidelines, F-IV, 4.18).